

Numbers on a Clothesline

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“A generating function is a clothesline on which we hang up a sequence of numbers for display.”
Herbert Wilf, *generatingfunctionology*

1 Introduction

In mathematics, one often deals with structures that can be mapped to the natural numbers; these may be in the form of sequences or sets of things, to give some examples. Generating functions are a powerful way to work with such structures, providing an algebraic lens with which to view combinatorial or number-theoretic objects. Indeed, we will see that this is where their power lies, as the ability to swap between algebraic and combinatorial thinking provides insights that are tremendously difficult to reach otherwise.

2 A Look at the Binomial Theorem

Theorem. The expression $(x + y)^n$ can be written as

$$\sum_{k=1}^n \binom{n}{k} x^k y^{n-k}.$$

Proof. For each factor in the expression $(x + y)^n$, we can choose whether the product contributes an x or a y ; making this choice for every factor generates some term of the form $x^j y^{n-j}$, depending on how many x s and y s were chosen.

Now, consider the coefficient of the $x^k y^{n-k}$ term. We want to choose k of the $x + y$ factors to take an x from, and fix the other $n - k$ factors to contribute a y , since this is the only way to get an $x^k y^{n-k}$ term. There are $\binom{n}{k}$ ways to do this, which means the product generates $\binom{n}{k}$ terms of the form $x^k y^{n-k}$, i.e., the coefficient of $x^k y^{n-k}$ is $\binom{n}{k}$, as desired.

The binomial theorem by itself is already a very useful theorem, and one that will feature prominently later. However, there's something more to be said about the concept shown here. Specifically, let's slightly modify the product to look like

$$\prod_{i=1}^n x_i + 1 = (x_1 + 1)(x_2 + 1) \cdots (x_n + 1).$$

What we have done here is set $y = 1$ and then given a label to each of the factors. Before we do anything, the first question we should think about is, what does the product look like when expanded? (Try it yourself!)

In the case of $n = 3$,

$$(x_1 + 1)(x_2 + 1)(x_3 + 1) = x_1x_2x_3 + x_1x_2 + x_2x_3 + x_1x_3 + x_1 + x_2 + x_3 + 1.$$

The interesting observation here is that each subset of $\{x_1, x_2, x_3\}$ appears within this sum exactly once: $\{x_1, x_2\}$ appears as x_1x_2 , $\{x_2\}$ as x_2 , and so on.

Now, how is this related to the binomial theorem? Let's take all the terms that have k x s, ignoring the numbers assigned to the individual x s. How many of these are there? Well, from the same argument, there are $\binom{n}{k}$ of these terms (which makes sense, considering we're counting subsets). This means that, in a sense, $(x + y)^n$ was counting precisely the same thing: the number of subsets of an n -element set.

The crucial idea captured here is that the coefficients of $(x + y)^n$ are **filtering the subsets** of $\{1, 2, \dots, n\}$ by the number of elements they have. This is the guiding principle behind generating functions as a whole: *Generating functions are algebraic objects that filter information.*

3 Counting with Functions, Part 1

This is the “classical example” of a combinatorial problem that's really hard to solve without a generating function.

Problem. Let $n = pk$ for an odd prime p and positive integer k . Given a set $S = \{1, 2, \dots, n\}$, find the number of subsets of S which have a sum of elements divisible by p .

Solution. First, what makes this a hard problem? The key difficulty is that normally, when dealing with the sum of a set's elements, one looks at more *global* aspects of the setup, such as the sum of all elements within subsets of a fixed size. Controlling the sum of a particular subset's elements is tricky with standard combinatorial methods.

This is where generating functions come in. Rather than filtering by the number of elements, we instead want to filter by the sum of the elements. This motivates the function

$$F(x) := (x + 1)(x^2 + 1)(x^3 + 1) \cdots (x^n + 1).$$

When expanded, the factor $x^j + 1$ either contributes 0 or j to each term, so each the degree of each term corresponds to the sum of the elements of the subset that generated this term. As before, the degree of each term is what allows us to filter for certain information. Now, we need some way to

get at the terms with degrees that are multiples of p .

The key insight to finish here is that, if we take $\zeta \neq 1$ to be a complex number with $\zeta^p = 1$, then

$$1 + \zeta + \zeta^2 + \cdots + \zeta^{p-1} = 0,$$

since $\zeta^p - 1 = (\zeta - 1)(1 + \zeta + \zeta^2 + \cdots + \zeta^{p-1}) = 0$. Using this, one can determine that

$$\frac{F(1) + F(\zeta) + F(\zeta^2) + \cdots + F(\zeta^{p-1})}{p}$$

is the answer, which turns out to be

$$\frac{2^{pk} + (p-1)2^k}{p},$$

with the computation details left as an exercise to the interested reader.

3.1 Practice Problems

1. A fair 6-sided die is rolled thrice. What is the probability that the sum of the three rolled numbers is 13?
2. A 6-sided die is rigged so that the probability of rolling k is $\frac{k}{21}$. If the die is rolled thrice, what is the probability that the sum of the rolled numbers is 13?

4 The Algebraic View

As we've seen, generating functions can be very powerful tools for filtering information about combinatorial setups where the objects can be partitioned into groups corresponding to integers. So far, however, our use of these objects has been limited to scenarios with finitely many objects. The theory we will develop in this section allows us to talk about *infinite* sequences as purely algebraic objects.

4.1 Ordinary Generating Functions

For a sequence $a_0, a_1, \dots$, denoted $(a_n)_{n \geq 0}$, we define its **ordinary generating function** (OGF) to be

$$A(x) := \sum_{k=0}^{\infty} a_k x^k.$$

In technical terms, this is a power series whose x^i coefficient is a_i . We attach the label “power series” rather than “polynomial” since A need not have a finite degree. Importantly, two OGFs are the same if and only if their corresponding sequences are identical.

Exercise. Let $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$ have OGFs $A(x)$ and $B(x)$, respectively. Verify the following:

- The OGF for the sequence $(a_n + b_n)_{n \geq 0}$ is $A(x) + B(x)$.
- The OGF $x^k A(x)$ encodes the sequence

$$(\underbrace{0, \dots, 0}_k, a_0, a_1, \dots).$$

- The OGF $A(x)B(x)$ encodes the sequence whose n th term is

$$\sum_{k=0}^n a_k b_{n-k}.$$

- The OGF $xA'(x)$ encodes $(na_n)_{n \geq 0}$.

The above properties allow us to combine sequences in interesting ways. In addition to this, we have the simple geometric series

$$\sum_{k \geq 0} x^k = \frac{1}{1-x},$$

which perhaps the most useful identity in manipulating OGFs. (Question: What is the derivative of this expression, and what is the corresponding power series?)

One of the most useful contexts in which one finds generating functions is the solving of recurrence relations.

Problem (Binet's Formula). Let $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$ (i.e. f_n is the n th Fibonacci number). Then,

$$f_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}.$$

Proof. Consider the OGF $F(x)$ of (f_n) . Then,

$$\begin{aligned} F(x) &= \sum_{n \geq 0} f_n x^n \\ &= \sum_{n \geq 2} (f_{n-1} + f_{n-2}) x^n + f_1 x + f_0 \\ &= x \sum_{n \geq 0} f_n x^n + x^2 \sum_{n \geq 0} f_n x^n + f_1 x + f_0 \\ &= (x + x^2)F(x) + x. \end{aligned}$$

Using partial fraction decomposition,

$$F(x) = \frac{x}{1-x-x^2} = \frac{1}{(1-r_1x)\sqrt{5}} - \frac{1}{(1-r_2x)\sqrt{5}},$$

where $r_1 = \frac{1+\sqrt{5}}{2}$ and $r_2 = \frac{1-\sqrt{5}}{2}$. Using the geometric series formula,

$$F(x) = \sum_{n \geq 0} \frac{(r_1^n - r_2^n)x^n}{\sqrt{5}}$$

which gives the claimed formula for f_n .

4.1.1 Exercises

1. Given any second order linear recurrence $a_n = Pa_{n-1} + Qa_{n-2}$ with $a_0 = c_0$ and $a_1 = c_1$, compute a_k in terms of P , Q , c_0 , c_1 , and k . (Be careful about the case where $x^2 - Px - Q$ has a double root!) Do you see how to extend this to a third order recurrence? An m th order recurrence?

2. Prove that

$$\frac{x^m}{(1-x)^{m+1}} = \binom{k}{m} x^k.$$

3. Recall the binomial theorem from earlier. In the language of generating functions, we can say that $(x+1)^n$ is the OGF for the sequence $a_k = \binom{n}{k}$. However, this can be generalized: in particular, the power series expansion of $(x+1)^r$, for any *real* r , is the OGF for $b_k = \binom{r}{k}$. Find a formula for b_k .

4.2 Exponential Generating Functions

The **exponential generating function** (EGF) of $(a_n)_{n \geq 0}$ is

$$\hat{A}(x) := \sum_{k=0}^{\infty} \frac{a_k}{k!} x^k.$$

Note: The hat above the A is not standard notation, but is used here as a reminder that this is not an OGF.

Much like the crucial identity for manipulating OGFs is

$$\sum_{k \geq 0} x^k = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x},$$

the corresponding identity for EGFs is

$$\sum_{k \geq 0} \frac{x^k}{k!} = \frac{1}{0!} + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = e^x.$$

4.2.1 Exercises

1. Show that...

- the EGF for the sequence $0, 1, 2, 3, \dots$ is xe^x .
- the EGF for the sequence $0, 1^2, 2^2, 3^2, \dots$ is $(x+x^2)e^x$.
- the EGF for the sequence $0!, 1!, 2!, 3!, \dots$ is $\frac{1}{1-x}$.
- the EGF for the sequence $\binom{0}{k}, \binom{1}{k}, \binom{2}{k}, \binom{3}{k}, \dots$ is $\frac{x^k}{k!} e^x$.

2. Let $A(x)$ and $B(x)$ be the EGFs of $(a_n)_{n \geq 0}$ and $(b_n)_{n \geq 0}$. Let $A(x)B(x)$ be the generating function for $(c_n)_{n \geq 0}$. Show that

$$c_n = \sum_{k \geq 0} \binom{n}{k} a_k b_{n-k}.$$

Manipulating EGFs is similar to manipulating OGFs, but with a slightly different flavor; there is not much new ground here.

4.3 Digression: What Does x Represent?

Often when working with a function, the point is to somehow map inputs to outputs in a meaningful way. Specifically, both the input and the output have some inherent meaning in the context of the problem.

In this sense, generating functions are not so much functions as they are an organizational tool. In our examples so far, generating functions have been used to sort items by some assigned index. This utility is the reason for the existence of x : it is a dummy variable which, due to its algebraic properties, is perfectly suited for such organizational structure.

Of course, one can derive information by inputting values into a generating function; this is what we did with the sum-of-subsets problem earlier. This is not, however, the goal of a generating function; x still does not have a meaning.

4.4 The Snake Oil Method

The Snake Oil Method is an algebraic trick for evaluating difficult summations. The steps are as follows:

- For some sum $a_n = \sum_k f(k, n)$, consider its OGF

$$A(x) := \sum_{n=0}^{\infty} \left[\sum_k f(k, n) x^n \right].$$

- Swap the order of summation: instead of summing over k , we instead will sum over n , with the hope that this is an easier sum to evaluate. In particular,

$$A(x) = \sum_k \left[\sum_{n \geq 0} f(k, n) x^n \right]$$

- Evaluate $A(x)$, and extract a_n from the coefficients.

The guiding principle behind this trick is that, since addition is commutative, we can swap the order of summation, potentially getting an easier sum.

The power of the Snake Oil Method is best illustrated with an example.

Problem. Evaluate

$$a_n = \sum_{k \geq 0} \binom{k}{n-k}.$$

Solution. Applying Snake Oil, we consider

$$A(x) := \sum_{n \geq 0} a_n = \sum_{n \geq 0} \sum_{k \geq 0} \binom{k}{n-k} x^n.$$

Then, swapping order of summation gives

$$\begin{aligned}
A(x) &= \sum_{k \geq 0} \sum_{n \geq 0} \binom{k}{n-k} x^n \\
&= \sum_{k \geq 0} x^k \sum_{n \geq 0} \binom{k}{n-k} x^{n-k} \\
&= \sum_{k \geq 0} x^k \sum_{r \geq 0} \binom{k}{r} x^r \\
&= \sum_{k \geq 0} x^k (x+1)^k \\
&= \sum_{k \geq 0} (x+x^2)^k \\
&= \frac{1}{1-x-x^2}
\end{aligned}$$

where $r = n-k$. As it turns out, this is (almost) the generating function for the Fibonacci numbers! Thus, the coefficient of x^n in $A(x)$ is $a_n = f_{n+1}$, the $(n+1)$ th Fibonacci number.

Here's another, more convoluted example.

Problem. Prove that

$$\sum_k \binom{m}{k} \binom{n+k}{m} = \sum_k \binom{m}{k} \binom{n}{k} 2^k.$$

Solution. We can use Snake Oil here, too, by showing that both sides have the same OGF. In particular, the generating function for the left side is

$$\begin{aligned}
A(x) &:= \sum_{n \geq 0} \sum_k \binom{m}{k} \binom{n+k}{m} x^n \\
&= \sum_k \sum_{n \geq 0} \binom{m}{k} \binom{n+k}{m} x^n \\
&= \sum_k \binom{m}{k} x^{-k} \sum_{n \geq 0} \binom{n+k}{m} x^{n+k} \\
&= \sum_k \binom{m}{k} x^{-k} \frac{x^m}{(1-x)^{m+1}} \\
&= \frac{x^m}{(1-x)^{m+1}} \cdot \left(1 + \frac{1}{x}\right)^m \\
&= \frac{(1+x)^m}{(1-x)^{m+1}}.
\end{aligned}$$

Doing the same thing on the right hand side gives

$$\begin{aligned}
B(x) &:= \sum_{n \geq 0} \sum_k \binom{m}{k} \binom{n}{k} 2^k x^n \\
&= \sum_k \binom{m}{k} 2^k \sum_{n \geq 0} \binom{n}{k} x^n \\
&= \sum_k \binom{m}{k} \frac{2^k \cdot x^k}{(1-x)^{k+1}} \\
&= \frac{1}{1-x} \sum_k \binom{m}{k} \left(\frac{2x}{1-x} \right)^k \\
&= \frac{1}{1-x} \left(1 + \frac{2x}{1-x} \right)^m \\
&= \frac{(1+x)^m}{(1-x)^{m+1}}.
\end{aligned}$$

Since $A(x) \equiv B(x)$, their corresponding sequences are the same, which is sufficient to prove the problem statement.

4.4.1 Practice Problems

1. For $n \geq 0$, compute

$$\sum_{a+b=n} \binom{2a}{a} \binom{2b}{b}.$$

2. For $m, n \geq 1$, compute

$$\sum_{k \geq 0} \binom{n+k}{m+2k} \binom{2k}{k} \frac{(-1)^k}{k+1}.$$

You may need the identity

$$\sum_{k \geq 0} \frac{1}{k+1} \binom{2k}{k} = \frac{1 - \sqrt{1-4x}}{2x}.$$

5 Counting with Functions, Part 2

5.1 Choosing Without Order

Generating functions are, as mentioned earlier, a way to filter information about a set of objects. The ideas in the previous section further ameliorate our ability to do this, with our newfound ability to talk about infinite sequences.

Operations between ordinary generating functions reflect the structure of choosing things without order. For example:

Problem. In how many ways can we fill a bag with n fruits subject to the following constraints?

- The number of apples must be even.

- The number of bananas must be a multiple of 5.
- There can be at most four oranges.
- There can be at most one pear.

Solution. Again, generating functions filter information. What is the mechanism for filtering here? Well, the obvious sequence we want is T_n , the number of ways to choose n fruits. This motivates filtering by the number of fruits.

Consider the function

$$A(x) := 1 + x^2 + x^4 + \cdots = \frac{1}{1 - x^2}.$$

This encodes the sequence $(1, 0, 1, 0, 1, \dots)$, which is the number of ways to choose n apples. Similarly,

$$B(x) := 1 + x^5 + x^{10} + \cdots = \frac{1}{1 - x^5}$$

encodes the number of ways to choose n bananas.

We also can only choose 0, 1, 2, 3, or 4 oranges, so

$$O(x) := 1 + x + x^2 + x^3 + x^4 = \frac{x^5 - 1}{x - 1},$$

and we can only choose 0 or 1 pears, so

$$P(x) := 1 + x = \frac{x^2 - 1}{x - 1}.$$

The important part is the following theorem:

Theorem. Let $R(x)$ encode r_n , the number of ways to choose n items from $\mathcal{R}$ and $S(x)$ encode s_n , the number of ways to choose n items from $\mathcal{S}$. If $\mathcal{R}$ and $\mathcal{S}$ are disjoint, then $R(x)S(x)$ is the OGF for c_n , the number of ways to choose n items from $\mathcal{R} \sqcup \mathcal{S}$.

Proof. Suppose we take j items from $\mathcal{R}$. We have r_j ways to do this. Then, we must take $n - j$ items from $\mathcal{S}$ to get n total items. This gives a total of $r_j r_{n-j}$ ways to choose n items, j of which are in $\mathcal{R}$. Summing over $j \in \{0, 1, \dots, n\}$ gives

$$c_n = r_0 s_n + r_1 s_{n-1} + \cdots + r_{n-1} s_1 + r_n s_0,$$

which is precisely the x^n coefficient of $R(x)S(x)$.

This theorem can, of course, be extended to more than two sets. In our case, applying it shows us that the OGF for t_n is

$$T(x) = A(x)B(x)O(x)P(x) = \frac{1}{1 - x^2} \cdot \frac{1}{1 - x^5} \cdot \frac{x^5 - 1}{x - 1} \cdot \frac{x^2 - 1}{x - 1} = \frac{1}{(1 - x)^2},$$

which we find, upon differentiating $\frac{1}{1-x}$, turns out to encode the sequence $t_n = n + 1$.

Exercise. Make your own problem! Try imposing arbitrary restrictions on the fruits and finding the corresponding generating function for t_n .

Here's another nice example!

Problem. To *triangulate* a k -sided polygon, we draw $k - 2$ non-intersecting diagonals, forming $k - 2$ triangles. How many ways are there to triangulate a k -sided polygon?

Solution. The key insight is that we can generate a recursion due to the inherent self-similarity of the setup. Let C_{k-2} be the number of triangulations of a k -gon.

Label the vertices $1, 2, \dots, k$ in order, and let (a, b) denote the segment connecting a and b . Then, $(1, k)$ must be a part of some triangle in this triangulation, whose third vertex is at i . Of course, $2 \leq i \leq k - 1$. Segments $(1, i)$ and (k, i) create two smaller polygons: one with i sides and one with $k + 1 - i$ sides. Hence, the number of ways to triangulate the new polygons is $C_{i-2}C_{k-1-i}$.

Summing over all i , we get that

$$C_{k-2} = \sum_{i=2}^{k-1} C_{i-2}C_{k-1-i},$$

or equivalently,

$$C_k = \sum_{i=0}^{k-1} C_i C_{k-i-1}$$

after re-indexing.

Before continuing, it's worth looking at the structure of this recurrence: importantly, the expression for C_k looks a lot like the result of squaring an OGF, since it is just

$$\sum_{i=0}^{k-1} a_i b_{k-1-i}$$

where we take both (a_k) and (b_k) to be the sequence (C_k) .

This is the final push that we need, since, taking $C(x)$ to be the OGF of (C_k) gives us the identity

$$G(x)^2 = 1 + xG(x)^2.$$

Funnily enough, this is a quadratic in $G(x)$, whose solutions are

$$G(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x}.$$

We now only need to decide which of the solutions is correct. We get this by taking $x = 0$, since $G(0) = C_0 = 1$. Analyzing both solutions shows us that, in fact,

$$G(x) = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

Exercise. The generalized binomial theorem says that

$$(1+x)^r = \sum_{n \geq 0} \binom{r}{n} x^n$$

for any real r . Use this to find a closed form for C_k . You will need the fact that

$$\binom{r}{n} = \frac{r(r-1) \cdots (r-n+1)}{n!}.$$

Fun fact: The numbers of the sequence (C_k) are called the **Catalan numbers**.

5.2 Choosing With Order

Just as OGFs are good for unordered choosing, EGFs work well for ordered choosing. This is due to the following theorem:

Theorem. Consider n objects in a line. Let a_k be the number of ways to give property α to a sequence of k objects, and b_k be the number of ways to give property β to a sequence of k objects. Let c_n be the number of ways to assign exactly one of properties α and β to the n objects. Then, if $A(x)$ and $B(x)$ are the EGFs of a_k and b_k , the function $A(x)B(x)$ is the EGF of (c_n) .

Solution. We know that the x^n coefficient of $A(x)B(x)$ is

$$\sum_{k \geq 0} \binom{n}{k} a_k b_{n-k}.$$

The goal is to show that this counts what we want. Fix k objects to give property α to; there are $\binom{n}{k}$ ways to choose these k . Then, there are $a_k b_{n-k}$ ways to assign the properties themselves. Thus, there are

$$\binom{n}{k} a_k b_{n-k}$$

ways to assign property α to any subset of k objects. Summing over all k gives the desired result.

Here's a pretty straightforward application of the theorem.

Problem. How many words are there in the alphabet $\{a, b, c, d\}$ such that there are an even number of as and bs (total) and at most 8 cs ?

Solution. Let t_n be the number of such words of length n . Our strategy is to construct an EGF for the number of as and bs together, and then multiply it by the EGFs for the number of cs and ds .

For the as and bs , take an even number $2m$. There are 2^{2m} ways to write a string of $2m$ as or bs , so the EGF for this would then be

$$F(x) := \sum_{k \geq 0} \frac{(2x)^{2k}}{(2k)!} = \frac{1}{2}(e^{2x} + e^{-2x}).$$

Since we have at most 8 cs , the EGF for cs is

$$C(x) := \sum_{k=0}^8 \frac{x^k}{k!}.$$

Finally, since ds are not restricted, the EGF for the number of ds is $D(x) := e^x$. The product is thus

$$\frac{1}{2}(e^{2x} + e^{-2x}) \cdot \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^8}{8!}\right) \cdot e^x.$$

To find a closed form, we rewrite the e^x terms to get

$$C(x) \sum_{k \geq 0} \frac{3^k + (-1)^k}{2} \cdot \frac{x^k}{k!},$$

so that the x^n coefficient is

$$\sum_{k=0}^8 \binom{n}{k} \left(\frac{3^k + (-1)^k}{2} \right),$$

where the summation only goes up to 8 because $C(x)$ only has nonzero coefficients up to x^8 , and this is the desired answer, however messy it may seem.

Exercise. Make your own problem, part 2! Perturb the setup, and find the corresponding generating function.

6 References

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